

Training Manual for Sorghum Production in Nigeria



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June 2021

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How to use this Manual

This manual is intended to provide farmers with easy-to-understand technical knowledge and cultural practices for Sorghum production without requiring advanced English literacy skills. This training manual can be utilized during pre-season crop planning and then again throughout the planting season and after harvest.

The facilitator's role is to read and understand this manual and then establish learning environments for farmers by providing opportunity for them to observe and understand differences, conduct basic tests and activities as described in the manual, and engage in conversations.

Even though this is a training guide, it is expected that the facilitator will listen to farmers and respect their knowledge, experiences, and perceptions, give farmers the confidence to share their knowledge and experiences, create appropriate conditions and activities from which farmers can learn, be responsive to farmers' needs, and be flexible in course organization.

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MODULE 1

General Conditions for Cultivation of Sorghum

Objectives of this module

- To enhance awareness of farmers on critical factors in Sorghum production.
- To create an understanding on importance of soil analysis prior to planting.
- To provide basic knowledge on how to identify good soil for sorghum production

IMPORTANCE OF SORGHUM

Sorghum is a traditional Nigerian crop that can be produced in locations where maize does not grow well. Sorghum is a grain that is higher in carbohydrates, vitamins, and minerals than maize. It contains more dietary fibre, as well as more protein, calcium, and iron.

Industrial demand of sorghum to produce composite flour by flour mills and the beverage companies have increased the demand for grain sorghum. Sorghum has higher micronutrients (Fe and Zn) compared to other cereals like maize, rice, and wheat, making it a healthier food. The grain has assumed commercial relevance, especially in the food and beverage industry as a valuable ingredient next to malted barley used in the industry. This is a guide to increase the productivity and production of sorghum.

It is native to Africa and has adapted to the continent's climate. It can tolerate periods of waterlogging and is drought resistant. It is one of the few viable food grains for most food insecure places, and it can be successfully grown in Nigeria's low rainfall areas (450-650mm per year).

Among the importance of sorghum

- Sorghum is high in iron and zinc and is processed into a variety of healthy dishes, including semi-leavened bread, couscous, dumplings, fermented and non-fermented porridges.
- Sorghum has been demonstrated to be the finest alternative to barley for lager beer brewing.
- In West Africa, a sorghum-based dye is used to dye leather crimson. Another product is sorghum starch.

Crop Description:

Sorghum is grasses between 50 cm and 6 m tall. Through breeding efforts, newer varieties now have 2-3 dwarf genes, resulting in a plant 2-4 feet tall and easier to harvest.

AGROCLIMATIC AND SOIL REQUIREMENTS

- The best time to plant is when there is sufficient water in the soil and the soil temperature is 15 °C or higher at a depth of 10 cm. Temperature plays an important role in growth and development after germination. A temperature of 27 to 30 °C is required for optimum growth and development.
- Sorghum is mostly grown in places with annual rainfall ranging from 300 to 750 millimetres.
- It is planted in locations where maize cannot be produced because the crop is drought tolerant.
- Sorghum can be cultivated in a variety of soil types, but the best yields come from deep, fertile, well-drained loamy soils. It, on the other hand, is tolerant to shallow soil and drought.
- Sorghum grows in different array of soils. However, it grows optimally in a well-drained heavy clay soil. It can also grow well on light and sandy soils.

pH

- It can withstand a range of soil acidities (from pH 5.5-7) and tolerates salinity better than maize.

SOIL SAMPLING AND ANALYSIS BEFORE PLANTING

Every area is different when it comes to soil types and nutrient contents in soil. Soil sampling and testing can show you the plant available nutrients and other soil chemical factors important for your crop production.

Nutrient levels in soil also vary from year to year, so it is important to perform soil sampling and testing prior to planting any new crop.

It is important for farmers to follow certain recommended steps for soil sampling and testing to develop a fertility management program.

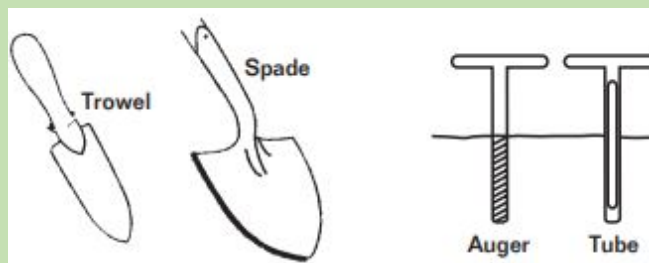


Figure 1: sampling tools for Sampling Soils for Fertilizer and Lime Recommendations and Frequency of Soil Sampling (Darryl Warncke (2013))

Sampling System

It is critical to collect soil samples from areas with similar features while collecting samples. As a result, you will need to divide the plot into portions with similar soil texture, slope, drainage, and cropping history. The samples can be taken in a random or zigzag manner; just make sure that the samples are evenly distributed across the entire sampled area.

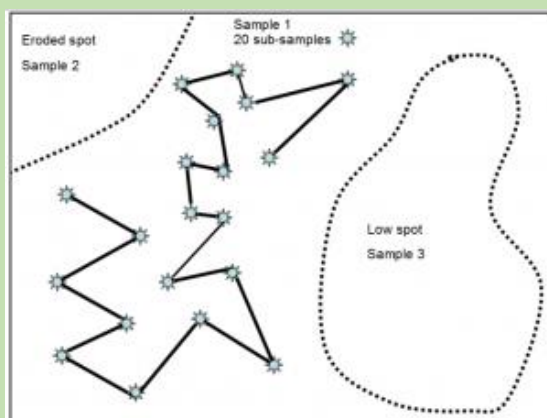


Figure 2: (by Cornell University, College of Agriculture and Life Science, 2012): Example of taking soil samples in the zigzag pattern. For each sampled area (spot) take multiple cores (sub-samples – marked as the blue sun on the photo) to get one homogenized sample

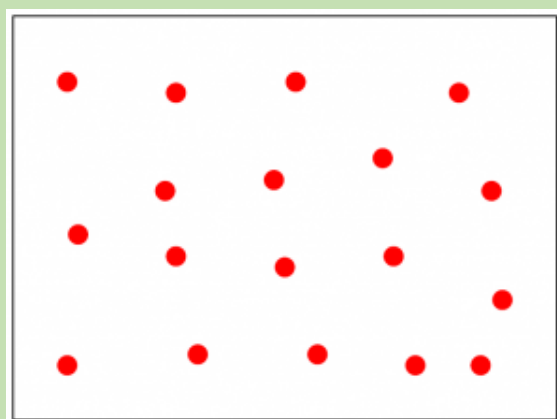


Figure 3:(by KIS, 2013): Example of taking soil sub-samples (cores) in the random pattern. Ensure uniform distribution of samples across one sampled area.

How to Take a Soil Sample from your farm

1. Selecting sampling spot

1. Select an appropriate sampling tool.
2. Make a sketch dividing the area into sampling areas.
3. Avoid wet or bare spots, and test on soil that has normal plant growth.
4. Remove the surface litter and take the correct depth of sample (15-30cm)
6. Make a “v” shape with a spade. Then remove a 1 in. slice from one side of the cut. Then remove a 1 in. slice from the middle.
7. Air dry the soil. Do not use heat! Mix your composite samples in a clean bucket. Place 1 pint of this into a nylon. Fill in the information on the nylon.
8. Send soil samples to testing agency in soil sample bag (reputable soil testing laboratory)

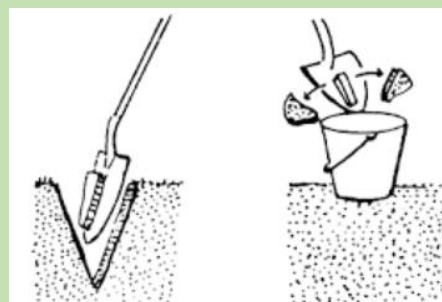


Figure 4: taking sample for soil analysis for Fertilizer and Lime Recommendations and Frequency of Soil Sampling (Darryl Warncke (2013)

Soil analysis

Soil analysis is an important tool for your farm since it identifies the inputs needed for optimal and cost-effective output. A proper soil test will guarantee that enough fertilizer is applied to meet the crop's needs while also taking use of the nutrients already available in the soil.



Figure 5: mixing composite samples (Darryl Warncke (2013)

Once you have collected your samples, make sure to get them into the lab as quickly as possible. Also, do not forget to specify exactly what kind of analysis you want. It is a good idea to get data on phosphorus (P), potassium (K), and nitrogen (N) levels, as well as soil pH, for Sorghum production site.



Figure 6: nylon for sample collection, (Darryl Warncke (2013))

Summary

Sorghum is a traditionally important crop in Nigeria that can be grown in marginal areas where maize may not thrive. Soil analysis is an important tool for your farm since it identifies the inputs needed for optimal and cost-effective output.

Q & A

What is the best soil for Sorghum production?

What are the important of soil analysis before planting?

Describe the procedure involve in soil sampling.

What are the importances of Sorghum in our society?

MODULE 2

PRE-PLANTING OPERATION

Objectives of this module

- To emphasize the importance of good land preparation
- To sensitize the farmers on good variety selection, seed treatment and seed requirement calculation
- To provide basic knowledge on how to measure and calculate the farm area
- To provide a simple way of carrying out seed germination test

Land Preparation

Clear the site of shrubs and stubble.
tractor.



Figure 8: Manual land Clearing
(Samuel Banister, 1912)

Alternatively, spray the field
with Glyphosate (Round-up)
to kill emerged weeds.



Figure 10: Land clearing using
Herbicides application, photo
adapted from shutter stock (2000)

Land can be prepared mechanically using



Figure 7: Mechanical land
ploughing (Samuel



Figure 9: manual land clearing using
hoe, photo adapted from
stovkphoto, 2004

Land can also be prepared
manually with the African hand-
hoe. Plow and harrow the field to
provide sufficient tilth for good
root growth.

Make ridges thereafter if desired.



Figure 11: Ridges made during OCP fertilizer trials on Soybean in Benue and Kaduna States, Nigeria (2019)

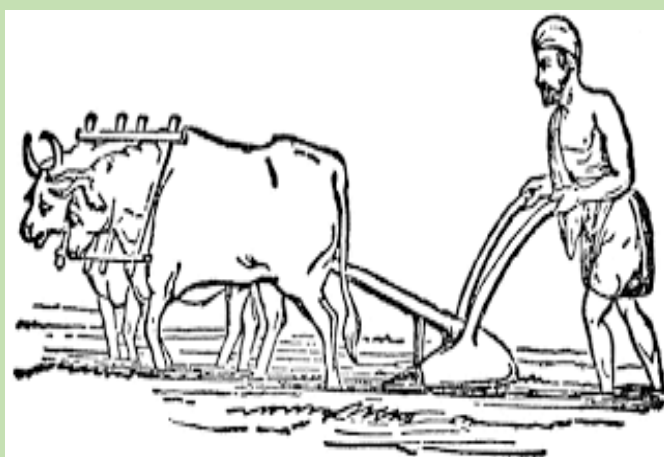


Figure 12: ploughing using animal (Samuel Banister, 1912)

Animal such as cattle, horse, and donkey are used in land preparation in most part of Nigeria.

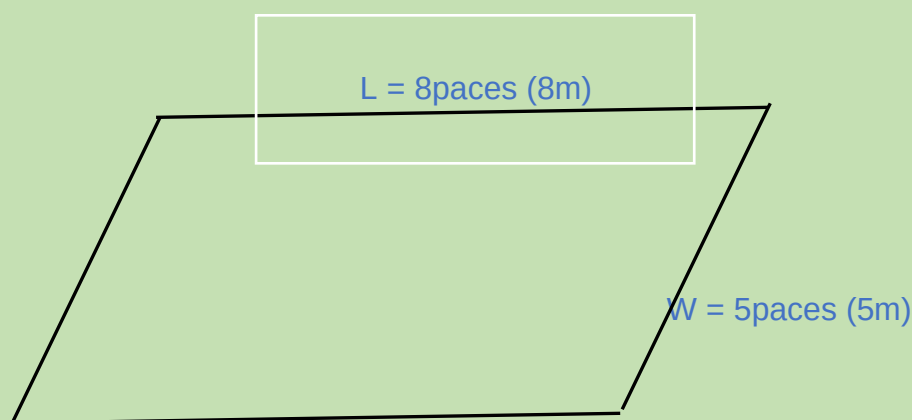
Estimating the Size of your land

Calculating the size of your plot is simple. All you need is a little patience and some basic math. Here is how to go about it.

You need to measure your plot because this will help you to buy the quantity of seed, pesticides and fertilizer needed and to plan properly.

To begin, pace the land's length (L) and breadth (W) and record the number of paces in each. One pace equals around 1 meter. Now divide the number of length paces by the number of width paces.

This will give you the square meters (m²) of the area. Look at the I for an example.



**Area of the
Land = L x W**

$$8 \times 5$$

$$40\text{m}^2$$

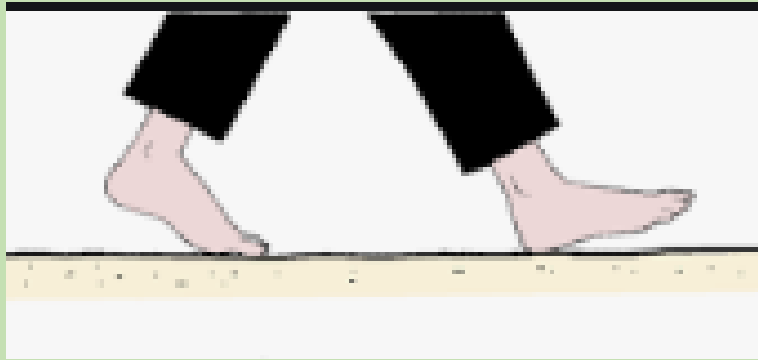
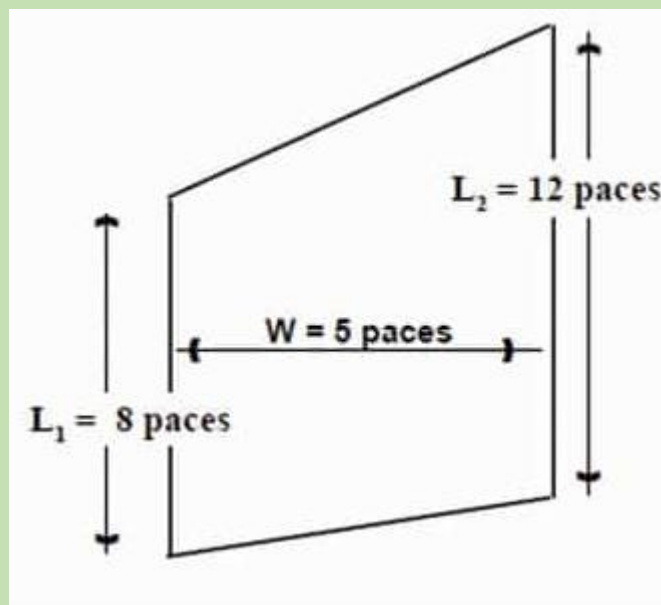


Figure 13: Pace measurement (FAO, 2005)

The above illustration works if your land is rectangle in nature.

If your land has a constant length but varying width, or a constant width but varying length (see the second illustration). You use a slightly different method to calculate the area.



Area of the Plot

$$= \frac{L_1 + L_2}{2}$$

$\times W$

$$\text{Area of the Plot} = \frac{8m + 12m}{2} \times 5m$$

$$\text{Area of the Plot} = \frac{20m}{2} \times 5m$$

$$= 10m \times 5m$$

$$= 50m^2$$

Alternatively

You can take the measurement of

your land using measuring tape

Remember One hectare is 100m X 100m.



You measure		For plot size you multiply	
Length of your plot in meters	Width of your plot in meters	Length with Width	
		=	m ²

You divide the size in m² by 10,000 to know the size in hectare.

Register here the size of your field.



Selection of Variety

Factors to consider when selecting varieties.

When choosing a variety to grow in your farming area some of the factors to consider include:

- The growing period (days to maturity),
- Yield potential,
- Susceptibility to bird damage/losses,
- Pest and disease resistance.

VARIETIES

Improved varieties are developed by IAR.

Short-season sorghum - the rainfall is less than 600 mm per annum.

medium maturing sorghum varieties - rainfall requirement ranges between 600-1800mm per annum, with about seven months of rain.

late-maturing sorghum varieties – over 180 days of rainfall.

Seeds are critical to any crop production. Good seeds of high yielding varieties will give a better yield and higher returns. Some of the improved varieties of sorghum are listed in the table below.

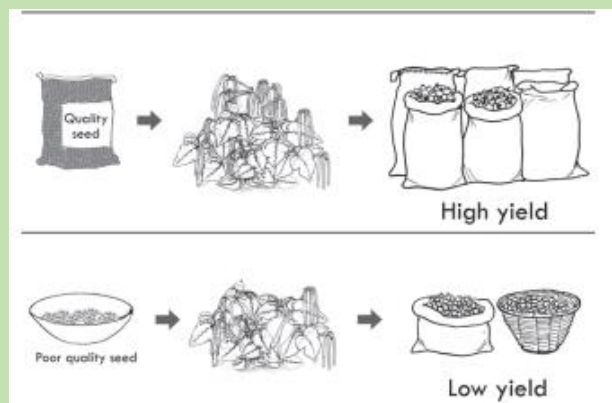


Figure 14: adapted from Seed production training guide by MUZARD and NACRRI, 2002

Table 1: Improved maturing sorghum varieties released by IAR Samaru

Release Name	New Name	Characteristics
KSV4(BES)	SAMSORG-3	Shot season type, maturity period 95-105 days (early) semi-dwarf. Resistant/tolerant to striga. Seed colour cream. Potential yields 1.5-2.5 t ha ⁻¹ .
KSV 11	SAMSORG-5	Shot season, maturity period 95-105 days (early) Tolerant to striga. dwarf type. Seed colour white. Potential yield,1.5-2.5 t ha ⁻¹
KSV 12	SAMSORG-6	Shot season, maturity period 95-105 days (early). Tolerant to striga. Semidwarf. Potential yield 1.5-2.5 t ha ⁻¹ , Seed colour cream.
ICSV4000	SAMSORG-40	Shot season, maturity period 95-100 days (early) Tolerant to striga. Semidwarf. Potential yield 2.5-3.5 t ha ⁻¹ , Seed colour cream. Use for malt production and brewing.
SSV98001	SAMSORG 42	Tall 2-3 m, Open panicles, white seeded, late maturing 105-120 days, Potential yield 2.5-3.0 t ha ⁻¹ , Tolerant to most leaf diseases, except Anthracnose
SSV98002	SAMSORG 43	Tall 2.7-2.8 m, Open panicles, late maturing 105-120 days, Potential yield 2.5-3.0 t ha ⁻¹ , Tolerant to most leaf diseases.
SSV20042	SAMSORG 44	Tall 1.9-2.0 m, Compact panicles, White seeded, Early maturing 95-100
KSV 3 (SK 5912)	SAMSORG-17	Long season, Semi-tall. Tolerant to striga. Maturity period 165-175 days. Potential yield 2.5- 3.5 t ha ⁻¹ , Seed colour yellow. For Industrial use especially for brewing, Livestock, confectionery.
SSV2 (FBL)	SAMSORG-16	Long season. Tall, Farafara, Maturity period 165-175 days. Potential yields 2.5-3.5 t ha ⁻¹ . Seed colour white. Use by Industries especially for brewing

Source: Idris Sanusi 2020 (Sorghum, production, processing and utilization)

Germination test

Prior to sowing, seed should be stored in a cool, dry, and rodent-free environment. To avoid a poor stand due to damaged seeds, a germination test should be performed at least 10 days before planting.

The steps below should be followed to conduct a germination test.

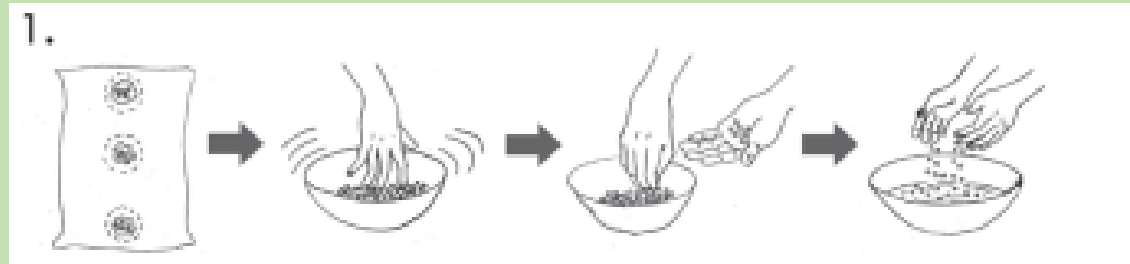


Figure 15: seed selection for germination test. Photo adapted from training guide by Irish aid 2000

- ❖ Take representative seed samples from the bag's top, center, and bottom.
- ❖ To use for the test, combine the sample seeds and count 100 seeds (for small seed available, farmers can count 20 seeds).
- ❖ Place the seeds in a water-filled jar for 24 hours.
- ❖ Drain the water and wrap the seeds in the saturated cotton towel to make a bag that will keep the seeds.
- ❖ Tie a piece of stick to the cloth bag. Tilt the stick to encourage the cloth bag to drain. Water the cloth three times a day to keep it moist. Allow 48 hours for it to be knotted.
- ❖ After that, untie the cloth bag and count how many seeds have fully germinated (both the shoot and roots have emerged)

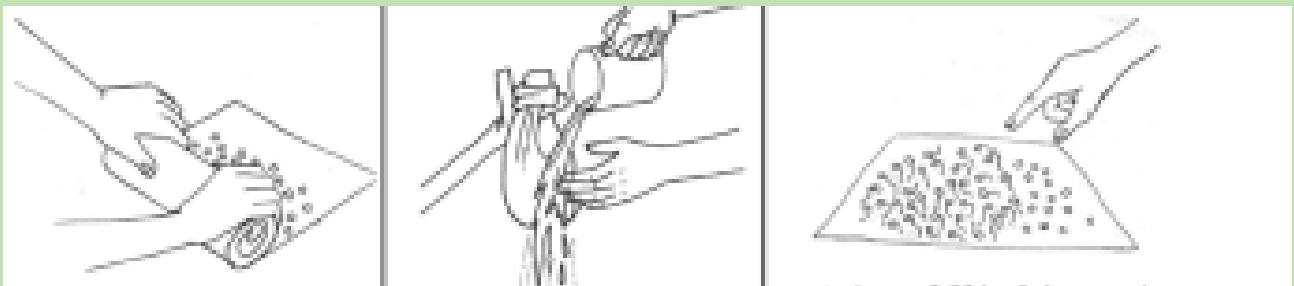


Figure 16: seed germination test. Photo adapted from training guide by Irish aid 2000

- ❖ Increase the seed rate at planting if the proportion is slightly less than 85 of 100 seeds. If the percentage is less than 70%, the seed should be discarded. The seed should not be used because it will provide low yields.
- ❖ Make a judgment based on the germination test results as shown in the table below.

Alternatively

- ❖ Pick 100 seeds at random and sow them at a depth of 2.5 to 5 cm in a prepared field or in a container at home (placed in the sun).
- ❖ Count the number of seedlings that emerge after 4 days.

- ❖ Make a judgment based on the germination test results as shown below.

Table 2: Decisions to take after a seed germination test.

Number of plants counted	Number seed to be planted
85 or more	2 seeds per hole
70- 84	3 seeds per hole
<70	Get better seed

Seed Requirement

The number of seed to be used will depend on the method of planting. If planting is done by hand in rows, you need 10-15kg of seeds of sorghum in one hectare of land.

Seed treatment

First, treat selected seeds with a mixture of insecticide and fungicide.

For example, Apron StarTM 42 WS (thiamethoxam 20 g/l + difenoconazole 2 g/l + metalaxyl-m 20 g/l) at the rate of one sachet per 4 kg seeds or any available seed dressing chemical before sowing.

Other products can be used:

PROCOT 40 WS (carbosulfan + carbendazim + metalaxyl-m. CALTHIO C 50 WS (thiram + chlorpyrifos-ethyl).

Advantages of seed treatment

- ❖ Prevents spread of plant diseases.
- ❖ Strong against critical early season diseases
- ❖ Protect seed from seed rot and seedling blight.
- ❖ Improves germination.
- ❖ Controls soil insects.
- ❖ Provides protection from storage insects.

Time to plant Sorghum.

Sorghum can be cultivated in different ways:

Rain-fed conditions - a great time frame for planting sorghum is between Mid-July and September. Drought (harmattan) would not have settled in by this time, meaning that the soil would neither be too moist or too dry — the perfect soil condition for sorghum farming in Nigeria.

Irrigated (Dry-season farming) – planting should be done between late December to early January. ---- water supply is at irrigation interval of 21 days.

Summary

- ❖ Use quality seed for better yields.
- ❖ Quality seed should be of uniform size, colour, and shape. It should also be free of foreign matter such as weed seed, chaff and should be pest and disease free.
- ❖ Before sowing, test seeds for viability and germination potential by conducting a quick germination test.
- ❖ Knowing the total plot size will help in determine the quantity of input needed.

Q & A

Give different method of land preparation?

What is the importance of germination test?

Describe the method of seed germination test.

MODULE 3

Planting Operation

Objectives of this module

- To sensitize the farmers on good sowing
- To provide basic knowledge on fertilizer requirement and application for Sorghum

PLANT SPACING, DEPTH AND POPULATION

Seeding rate and plant population

Seeding rate varies from 7 to 12 kg/ha depending on the different seed sizes as well as the desired plant population. The plant population varies from 130 000 to 150 000 plants per hectare. Poor seed quality, poor seedbed preparation, insufficient soil moisture, insects and diseases can result in a poor stand. The quantity of seed should therefore be increased to compensate for a poor stand. On the other hand, the population may be too high for the prevailing water and nutrient supply, if germination is good.

Planting depth

Sorghum has a small seed and should be planted shallow. A planting depth of 25 mm is satisfactory with sufficient water. Under drier conditions the seed should be planted deeper, but no more than 50 mm. Planting depth is also determined by soil type. On heavy soils, the planting depth should not be more than 25 mm, while on light soils the depth can be as much as 50 mm. It is important that the soil surrounding the seed is firm to ensure rapid absorption of water and therefore germination.

The recommended plant spacing is as follows:

Inter row: 60 to 70 cm for areas with moderate rainfall and 90cm for drier areas.

In-row: 15 to 20cm for areas with moderate rainfall and 30cm for drier areas.

Recommended plant population for sorghum is 130 000 to 150 000 plants per hectare.

PLANTING METHOD

Two methods of planting can be used, that is, hand planting and machine (planter) planting. When machine planted, sorghum is normally planted using maize planters. Adaptations should be made by using the correct planter plates.

Plant 5 seeds per hole and cover the seeds with 2-3 cm of clod-free soil.



Figure 17: use of direct seeder for planting sorghum photo adapted from fao 2012

THINNING

- At 2-3 weeks after germination or at the 3-4 leaf stage uproot the weakest & leave the 2 strongest plants per basin
- If the soil moisture is favourable, the uprooted plants can be transplanted.
- The resultant average number of plants per stand should be 2.
- High plant population can lead to poor crop growth, especially under drought conditions.



Figure 18: Thinning of sorghum photo adapted from Access agriculture 2010

FERTILIZER APPLICATION

- Ideally, the amount of fertilizer to apply should be determined through soil testing.
- However, as a rule, the recommended rate of basal fertilizer application for optimal yields is 200 kg/ha of NPK11:22:21+5%S+0.7%Zn+0.5%B.

Method of application

- Basal fertilizer should always be applied before or at planting, under the seed, making sure that it does not get into direct contact with the seed.
- Make a furrow of 5-7 cm deep. You will also use this furrow to plant Sorghum.
- Place the fertilizer in the furrow and cover with 2 cm of soil.
- Cover the fertilizer with a layer of soil (1-2 cm thick) before seeding.

Note

Apply 2 heaped beer bottle caps (equivalent of 200 kg/ha of NPK11:22:21+5%S+0.7%Zn+0.5%B) of basal fertilizer per basin.



TOP DRESSING

- A top dressing of 100 to 200 kg per ha of 28 - 34% N fertilizer is recommended and should be applied at 4-6 weeks from crop emergence date.
- Place the top-dressing fertilizer 5-10 cm from the plant and avoid getting the fertilizer into contact with the plant to avoid fertilizer burn.
- Top dressing must be done when the soil is moist, preferably soon after raining.
- Split application is recommended at 4 weeks and second application at 6 weeks from planting date.

Summary

- ❖ Sorghum has a small seed and should be planted shallow.
- ❖ Ideally, the amount of fertilizer to apply should be determined through soil testing.
- ❖ Seeding rate varies from 7 to 12 kg/ha depending on the different seed sizes as well as the desired plant population.

Question

What is the significance of fertilizer use in crop production?

What is the spacing recommended for Sorghum?

What is the amount of seed in mudu to use for a hectare of land?

What are the different methods of applying fertilizer for Sorghum?

MODULE 4

FIELD MANAGEMENT

Objectives of this module

- To sensitize the farmers on good post plant practices, weeding and pest & diseases management
- To provide basic knowledge on different types of herbicides/pesticides



Weed and weed control.

Weeds are undesired plants that grow in the field among the crops. Weeds battle for water, nutrients, space, and light with the plant. Weeds, like other plants, feed on and drink from the same soil that our crops do. Once these weeds are eradicated, crops and plant growth will improve, and crop yields will increase.

WHY AND WHEN SHOULD WEED MANAGEMENT BE DONE

- Inefficient weed control is one of the main causes of low yields in sorghum.
- Some weeds become alternative hosts of pests and diseases. They reduce profits by lowering the quality, quantity, yields and value of sorghum.
- Some weeds are parasitic and poisonous to sorghum.
- It is important to control weeds throughout, but especially in the early stages of crop growth.
- This can be done through use of herbicides, cultural methods, and hoe weeding.
- As a general recommendation, first weeding at 5 to 6 leaf stage, just before top dressing. The second weeding is done after 6 to 7 weeks.

C COMMON WEEDS IN SORGHUM



Figure 21: Yellow Nutsedge, FAO 2013



Figure 20: Bamuda grass, FAO 2013



Figure 19: Striga, FAO 2013

METHODS OF CONTROLLING WEEDS

Manual or Mechanical methods

- Weeds can be removed manually (hand weeding and use of hand tools) and mechanically (using mechanical implements).
- For sorghum effective weed control requires 2 to 3 weeding.
- Carry out the first weeding within 2 weeks after planting and the second as soon as weeds appear around 5–6 weeks after planting, depending on weed pressure.
- Avoid weeding immediately after it has rained as this would not offer effective weed control.

Cultural practices Practices.

- Proper seed placement, fertility management, planting date, and seeding rates can help establish a healthy, competitive sorghum crop.
- Ploughing during winter is an effective method of controlling weeds.

Chemical methods

- Chemicals formulated as liquids, granules or gases can be applied to kill germinating, growing weeds or seeds. Herbicides offer a cost-effective method of weed control if used properly.
- Application should be based on correct weed identification.
- The choice of herbicide, however, depends on the predominant weed species and the availability of the herbicide.
- Herbicides are available for pre-emergence or post emergence weed control depending on time of application.
- If herbicide is applied at planting, one weeding may be required at 5–6 weeks after planting depending on weed presence.

HERBICIDE APPLICATION

Types of herbicides

- Pre-plant: incorporate into the soil before planting.
- Pre-emergence: applied after planting before germination of the weed,
- Post emergence: applied after emergence of crop and weeds.

Table 3: Some Herbicides which can be used in sorghum.

Active ingredient	Trade Name	Timing	Application rates and details
Atranex	Atrazine	Pre and early post-emergence	3-4 litres per hectare Controls most annual grasses and broad leaf weeds. Not to be used on coarse soils. Long residual effect.
Metolachlor	Dual Magnum	Pre-emergence	2-3 litres per hectare Controls some annual broad leaf weed, grass weeds and yellow nutsedge
Bentazon	Basagran	Postemergence	2-4 litres per hectare Controls broad leaf weeds and suppresses yellow nutsedge. Apply at first 3 leaf stage and use higher rates and more water for sedge
Paraquat	Gramoxone	Post-emergence	1.5 - 2 litres per hectare for control of various broad leaf weeds and all grasses. Nonselective foliar applied herbicide
Glyphosate	Round Up	Post-emergence	4 -5 litres per hectare depending on weed size. Nonselective foliar applied herbicide. Active on annual and perennial weeds

Source: Weed control in sorghum (Akunbodu, 2005)

Safe use of Herbicides/pesticides

For safe use of pesticides/herbicides farmers should respect these rules

- ❖ Use pesticides/herbicides approved by Ministry of Agriculture and Rural Development of Nigeria.
- ❖ Cautious storage, use of pesticides and recycling of used packing material out of children's reach.
- ❖ Read the label of the pesticides to know the correct dosage and utilization or contact an extension officer for assistance.
- ❖ Wear protective clothes and mask.
- ❖ Maintain the sprayer well to ensure that the sprayed quantity is correct. This saves money and work time.
- ❖ During spraying, do not eat, drink, or smoke.



Figure 22:
protective

- ❖ Spraying is work for adult professionals.
- ❖ Wash the training materials thorough and store them.
- ❖ Bury or burn empty containers in a metallic barrel far from houses when there is no wind.
- ❖ After spraying, wash yourself and change your clothes.

PEST AND DISEASE MANAGEMENT

DISEASES

1. Head Smut

The entire head is replaced by a large smut gall covered by a thick white membrane. This is the most common disease in sorghum. The disease spores are easily carried by wind and rain, spreading the disease to other plants.

Management

- Use of resistant varieties and
- use of certified disease-free seed.
- Rouging smutted heads and burning them
- Crop rotation



Figure 23: Head smut, FAO 2012

2. Covered kernel Smut.

Symptoms include all kernels in affected smut head destroyed and replaced by a dark brown powdery mass.

Management

- Use of resistant varieties
- Use of certified disease-free seed.
- Rouging smutted heads and burning them - Crop rotation



Figure 24: Covered kernel Smut, FAO 2012

3. Leaf blight

Large elliptical spots up to 20 mm wide and 100 mm long, initially water soaked, but drying to straw-coloured spots with red, purple, or tan margins depending on the hybrid.

Management

- Rotation with non -susceptible crops.
- Use of resistant hybrids
- Seed dressing with Captan or Thiram.
- Use of foliar fungicides.



Figure 25: Leaf blight, FAO 2012

PESTS

Stalk borer

The most important pest of sorghum. Symptoms starts with small holes in leaves then in stems. Larvae then channel through the stem eating the central part.

Management

-Cultural methods include deep ploughing and destruction of stover to destroy overwintering larvae, rotation with legumes to interrupt the life cycle of stem borers and addition of nutrients to the soil.

-Chemical control include Carbaryl 5 dust; Carbaryl 85WP; Dipterex and Thionex; Kombat Carbaryl 85%: 2kg/ha; Dipterex 2.5%: 1-2kg/ha. Carbaryl is applied as full cover after emergence. Ensure penetration into the funnel.



Figure 26: Stalk borer, FAO 2012

African Army worm

Pests cut and chew causing extensive damage in sorghum. They also consume the lamina leaving midribs thereby reducing the photosynthetic area.

Management

Controlled by use of contact insecticides which include:

-Carbaryl 85WP -Karate 5EC apply as full cover spray



Figure 27: African Army worm, FAO 2012

Aphids

These usually appear during head emergence and flowering; they suck sap from the leaves.

Management

Systematic control after head emergence with chemicals such as malathion 85% and Dimethoate.



Figure 28: Aphids, FAO 2012

Birds:

The worst nightmare for the sorghum farmer and can cause significant damage and reduction in yield. These become a problem as the crop approaches maturity.

Management

- Bird scaring is the only effective way of minimizing bird damage, but community cooperation in planting dates may also help to spread the risk.
- Farmers are encouraged to harvest early to minimize bird damage.
- Clear possible habitats around the field
- Growing of sorghum by many farmers in same locality will reduce losses.

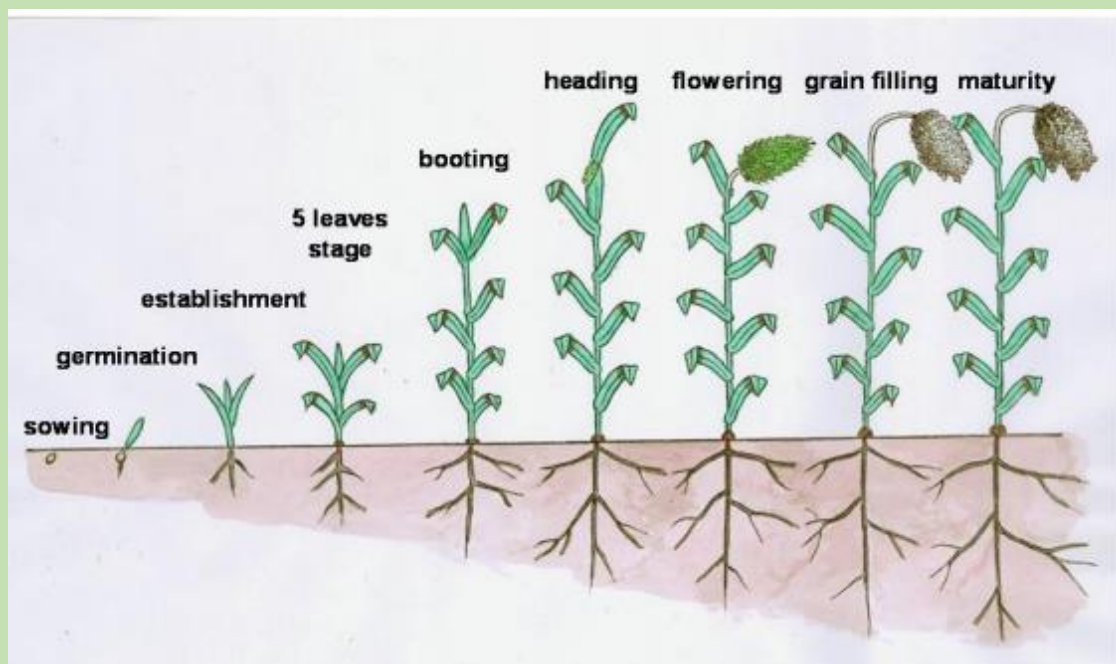


Figure 29: Growth and Development stages of Sorghum

Adapted: http://sanangelo.tamu.edu/programs/agronomy/publications/grain_sorghum/sorghum_plant_development.php

Stage 0: Initial emergence

3 to 10 days after sowing, the plant begins to emerge. Growth is influenced by soil temperature and moisture, planting depth, and seed germination ability during emergence.

Stage 1. Three–Leaf Stage

The three–leaf stage occurs when the collars of three leaves can be seen without dissecting the plant. Depending on the temperature, this stage may occur approximately 10 days after emergence.

The stage 2. is the five-leaf stage.

The five–leaf stage occurs three weeks after emergence, when the collars of five leaves may be seen without dissecting the plant. At this point, the root system is quickly developing. If not addressed, stresses like as weed competition, nutrient deficits, drought, or insect damage can drastically lower yields.

Stage 3. Growing Point Differentiation

At this stage, the growing point of the sorghum plant changes from vegetative to reproductive. Nutrient uptake is rapid and adequate supplies of nutrients and water are necessary to achieve maximum growth. This stage occurs approximately 30 days after emergence and is about one third of the time from planting to physiological maturity.

Stage4. Boot Stage

At this stage all the leaves are fully expanded, providing maximum leaf area and light interception. The head has reached full size and is encompassed by the flag-leaf sheath. Rapid growth and nutrient uptake continue. Stress from lack of moisture or via herbicide injury may prevent the head from emerging completely from the flag-leaf sheath preventing complete pollination at flowering.

Stage 5. Half Bloom

This stage is defined as when half of the plants in a field have started to bloom. Flowering progresses from the tip of the head downward over a period of 4 to 9 days. Severe moisture stress can result in poor head filling.

Stage 6. Soft Dough

At this stage, the grain has a dough-like consistency and grain fill is occurring rapidly. The lower leaves continue to senesce with 8 to 12 leaves remaining at this stage.

Stage 7. Hard Dough

By this stage approximately three-fourths of the grain dry weight has been attained. Nutrient uptake at this point is essentially complete. Severe moisture stress caused by an early freeze at this growth stage can result in light, chaffy grain.

Stage 8. Physiological Maturity

At this stage, the maximum total dry weight of the plant has been reached. This stage is marked by the appearance of a dark spot on the opposite side of the kernel from the embryo. Grain moisture at physiological maturity depends on the hybrid, with typical moisture ranging from 25% to 35%.

Summary

- Inefficient weed control is one of the main causes of low yields in sorghum.
- Proper seed placement, fertility management, planting date, and seeding rates can help establish a healthy, competitive sorghum crop.
- Bird scaring is the only effective way of minimizing bird damage, but community cooperation in planting dates may also help to spread the risk.

Question

Explain Post planting management involve in Sorghum production.

MODULE 5

Harvesting and Post-harvest activities

Objectives of this module

- To provide knowledge on harvest, processing, and proper storage of Sorghum

HARVESTING & POST HARVEST MANAGEMENT

HARVESTING SORGHUM

Harvesting is done when the crop reaches physiological maturity which is indicated by the following indicators:

- The spot where the grain attaches to the inflorescence turns from green to black.
- Yellowing and drying of leaves

Where birds are a serious problem, heads can be harvested at this stage and dried on floors or racks. When hand harvesting, the sorghum heads are cut. The heads are dried in heaps on the ground or threshing floor. If the entire plant is cut by hand, it should be stacked and left in the field to dry and mature for 10 to 14 days, and then threshed. A combine harvester can be used to harvest the sorghum. Hand harvested heads can be threshed by a stationary combine, hand beating or by driving cattle over the layers of sorghum heads.

YIELDS

Sorghum hybrid/improved variety yields range from 3 to 6 tonnes depending on the variety.

Transport

Sorghum is transported from harvest fields to threshing or drying facilities, then to storage facilities or gathering centres, processing facilities or larger central storage buildings, and ultimately to wholesalers or retailers for final sale.

Traditional transport system

Sorghum is transported by humans, donkeys, and horses in places where the road network is



Figure 30: Mature Sorghum photo adapted from science photo.com



Figure 31: woman harvesting sorghum women transporting harvested sorghum, source ICRISAT, 2004



Figure 32: women transporting harvested sorghum, source ICRISAT, 2004

inadequate, and agriculture is traditional. Tiny trucks, on the other hand, are used to deliver small quantities of beans that are often packaged in bags.



Figure 34: transport of farm produce using animal adapted from Sorghum production (FAO, 2012)



Figure 33: transport of farm produce using truck adapted from Sorghum production (FAO, 2012)



Figure 35: transport of farm produce using truck adapted from Sorghum production (FAO, 2012)

POST HARVEST MANAGEMENT FOR SORGHUM

Post-harvest options are generally all the activities that can be carried out after the harvesting of crops to convert it to use by man and animal. It can be classified into primary and secondary processing.

Major losses are incurred during field drying/harvesting, storage, and homestead drying. Harvesting, threshing, winnowing/cleaning, and storage are the critical farm level loss points for sorghum.

Drying Drying.

- Should target to reduce the moisture content to 12-12.5%
- Sorghum heads can be dried naturally by putting the heads on top of a black plastic/ concrete floor or traditional cribs using the sun.
- Artificial drying is done using threshers.



Figure 36: Drying harvested sorghum, photo adapted from Africa stock photo 2001

Threshing

This is the separation of grain from the heads. This is normally done manually using beating sticks or using grain shellers.



Figure 37: metal silo, Source: Institute of Agricultural Engineering, Ministry of Agriculture Mechanisation and



Figure 38: Threshing of Sorghum by beating. Source Marco et al., 2013

Storage

Sorghum best stored as whole grain and most importantly the seed stored at 12 to 13 % moisture or less. The objective of storage is to preserve as much as possible of the value of the grain for its intended future use. This means either retaining as high a proportion of viable seeds as possible for planting at the next harvest or preserving as much as possible of the food value of the grain for as long as possible.

Storage methods

The sorghum grain can be stored for future use in hermetic bags or Metal silos, which come in different sizes.



Figure 39: Hermetic bag, Source: (<https://www.rockefellerfo>)

Summary

Harvesting is done when the crop reaches physiological maturity which is indicated by the following indicators:

- The spot where the grain attaches to the inflorescence turns from green to black.
- Yellowing and drying of leaves

Major losses are incurred during field drying/harvesting, storage, and homestead drying.

Sorghum best stored as whole grain.

Question

Identify signs of physiological maturity for sorghum and when harvesting should practically be done.

Identify sorghum post-harvest loss points and how to address them.

MODULE 6

Farm Records

Farm record

Farm record is the documentation or the writing down of all the activities that are taking place on the farm.

Example of planting record sheet

Plant Date	Crop	Variety And the source	Total square meter planted
1/30/2021	Sorghum		10000

Monthly Labour Tracker records

Date:	Crop:	Bed Number(s):	Activity:	Time (to the 15 minute):	Paid labor/unpaid (e.g.yourself)

Monthly Harvest/Revenue record

S/N	Date	Crop	Harvest (ton/ha)	Quantity/ha	Quantity. sold	Price	Revenue

What is the importance of farm record?

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